IMT Doctrine v0.2.3-demo + 3 changes

Release v0.2.3-demo + 3 changes

Doctrine working group

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Warning

Demonstration repository. This site shows how the LandSAR IMT doctrine could be written, reviewed and published from Git. Structure follows the draft; text is short illustrative placeholder content, not agreed doctrine.

Cite as: IMT Doctrine, version v0.2.3-demo + 3 changes (commit f6d3a18).

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CHAPTER

ONE

PART 1 FOUNDATIONAL OVERVIEW

1.1 1.1 Introduction

LandSAR volunteers provide an independent, impartial and professional response in support of New Zealand's two Coordinating Authorities: New Zealand Police and the Rescue Coordination Centre New Zealand (*RCCNZ*). When a Coordinating Authority engages LandSAR, specialist Incident Management Teams (*IMT*s) manage the operation. Volunteers also provide search and rescue capability during civil defence emergencies.

Land search management has developed unevenly, drawing on many government and non-government sources, so practice varies across the country. This doctrine sets out the principles, core concepts and proven methods that govern how IMTs organise, train and operate. It is the single source of truth for "how we do things".

The doctrine is:

- **Integrated**: it fits within the legislative and operational requirements of the Coordinating Authorities.
- **Authoritative, not prescriptive**: it gives a decision-making framework while preserving the flexibility IMTs need.
- **Based on proven practice**: it captures lessons from past operations and reviews.
- **Grounded in search theory**: local, national and international.
- **The foundation for consistency**: shared terms, processes and expectations so groups, Police and partners work together seamlessly.
- **Built for continuous learning**: one place to add, amend and improve.

1.2 1.2 Purpose

To set a unified, nationally consistent approach to IMT response processes and practices. The doctrine:

- draws on input from IMTs nationwide
- brings together national and regional coordinators, tutors, group IMT leaders and Police SAR coordinators
- pairs a single source of truth with a feedback loop, so operational learnings update the document
- embeds its principles in IMT training and operational evaluation
- makes it simple to benchmark against international practice.

1.3 1.3 Scope

The doctrine covers the delivery of effective land search and rescue within an operational management context. It cascades from principles to fundamental concepts, then to operational application, guidelines, and the templates, checklists and forms that follow from them. Training material is built on the same foundation.

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It is not:

- a guide to every SAR sector function or agency activity
- a prescriptive, step-by-step operations manual
- a set of local standard operating procedures.

Note

Scope is still to be settled against the doctrine brief, for example whether notification and Coordinating Authority processes are in or out.

1.4 1.4 Assumptions and applicability

The doctrine consolidates current New Zealand practice from training, volunteer and Police operational experience, IMT tutors and national and international publications. It is deliberately not exhaustive.

It applies to all LandSAR IMT personnel and assumes endorsement by Police and RCCNZ. It rests on a **federation model**:

Who	Role
National LandSAR	Strategy, standards, quality assurance, data systems, national partnerships
Local groups	Local identity and relationships, readiness, front-line response
Coordinating Authorities	Statutory control of the search and rescue operation (*SAROP*)

1.5 1.5 Audience

The primary audience is volunteer IMT members. The doctrine also gives Police, RCCNZ and partner organisations a transparent view of how LandSAR manages operations, and other SAR organisations may use it as a reference.

1.6 1.6 Strategic context

New Zealand's terrain, weather, rivers, caves, alpine areas and visitor mobility demand local knowledge within a nationally coherent system. Coordination is split by jurisdiction:

Category	Coordinated by	Typical operations
I	New Zealand Police (locally)	Land, subterranean, river, lake and close-to-shore
II	RCCNZ (nationally)	Missing aircraft, distress beacons, offshore marine

The doctrine integrates established international search theory with local protocols and training. Joint Police, RCCNZ and LandSAR training is the key to a consistent approach.

The **keep it simple** principle applies throughout. Groups with high operational tempo can go further with extended search planning and advanced theory, provided their insights come back into the national doctrine.

Condensed from draft v0.1, Part 1.

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CHAPTER

TWO

PART 2 PRINCIPLES

The "why" behind what we do. These 15 principles guide decisions, support inter-agency work and drive continuous improvement. They sit at the top of the derivation hierarchy: every concept traces back to at least one of them.

Numbering follows the draft. The order and grouping of principles is still open for review.

2.1 PRN-01 Unified and consistent approach

PRN-01 · Principle · version 1

- **Supports:** *CON-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Every team, wherever it is, follows the same recognised practices and processes. Consistency makes large operations work, lets groups and Coordinating Authorities operate together, and gives a clear basis for national training, competencies and operational reviews.

Local groups bring resources and knowledge of terrain, community and weather. National doctrine, training, systems and assurance make that knowledge interoperable and scalable.

In short: set the principles once, apply them consistently everywhere.

Condensed from draft v0.1, Part 2, principle 1.

2.2 PRN-02 Adherence to a national management framework (CIMS)

PRN-02 · Principle · version 1

- **Supports:** *CON-04*, *CON-06*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

LandSAR incident management aligns with the *CIMS*, applied flexibly to the land SAR environment. Terms and acronyms follow the CIMS manual, with LandSAR-specific terms only where needed.

CIMS is a scalable management structure. It does not say what actions to take, what the priorities are, or how to reason about the subject, environment and resources. That comes from SAR knowledge and experience.

In short: CIMS is needed for multi-agency interoperability, but it is not a SAR manual.

Condensed from draft v0.1, Part 2, principle 2.

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2.3 PRN-03 Fostering partnerships and relationships

PRN-03 · Principle · version 1

- **Supports:** *CON-05*, *CON-06*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

LandSAR is a **support agency** to Police and *RCCNZ* under a memorandum of understanding. The Coordinating Authorities hold authority and legislative responsibility; LandSAR provides operational expertise, tactical advice, specialised people (including trained IMTs) and local knowledge. LandSAR also supports other agencies where appropriate, for example in civil defence emergencies.

Diverse IMTs, with varied backgrounds and perspectives, avoid groupthink and produce more robust strategies.

In short: what our partnerships look like to us.

Condensed from draft v0.1, Part 2, principle 3.

2.4 PRN-04 Importance of regional collaboration

PRN-04 · Principle · version 1

- **Supports:** *CON-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Groups meet proactively, share expertise and stay aligned with national standards, so they can call on each other quickly during operations. No team is isolated; all benefit from the collective experience of IMTs nationally.

In short: none of us is smarter than all of us.

Condensed from draft v0.1, Part 2, principle 4.

2.5 PRN-05 Proactive planning and readiness

PRN-05 · Principle · version 1

- **Supports:** *CON-03*, *CON-13*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

IMTs pre-plan for likely events in predictable or repeat locations, and practise those plans. Every operation should have a suitably trained and resourced IMT. Never assume an operation will be over quickly: set up the team and processes as if it will be protracted.

In short: pre-plan to improve efficiency, and resource the IMT properly.

Condensed from draft v0.1, Part 2, principle 5.

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2.6 PRN-06 Effective and supportive leadership

PRN-06 · Principle · version 1

- **Supports:** *CON-05*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Leadership matters inside the team (cohesion and clarity) and between teams (trust and coordination with other groups and agencies). Good leaders collaborate, communicate clearly, keep situational awareness and seek expertise when needed. They empower team members to contribute.

In short: leadership affects operational outcomes; develop good leaders inside and outside the organisation.

Condensed from draft v0.1, Part 2, principle 6.

2.7 PRN-07 The critical role of communications

PRN-07 · Principle · version 1

- **Supports:** *CON-10*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The overriding communications principle is **information superiority**: one reliable, integrated flow of intelligence to the IMT, giving shared situational awareness and enabling safe, coordinated deployment by all agencies.

In short: without intelligence coming in, the IMT cannot predict, plan, resource or oversee effectively.

Condensed from draft v0.1, Part 2, principle 7.

2.8 PRN-08 Evidence-based and intelligence-led operations

PRN-08 · Principle · version 1

- **Supports:** *CON-08*, *CON-09*, *CON-11*, *CON-17*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Operations rest on evidence and are guided by intelligence. Analytical methods give structure, complemented by the knowledge and intuition of experienced people. The same evidence-based approach applies to building doctrine, training, planning and deployment.

In short: use evidence and intelligence to support or refute our thinking and actions.

Condensed from draft v0.1, Part 2, principle 8.

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2.9 PRN-09 Search urgency

PRN-09 · Principle · version 1

- **Supports:** *CON-04*, *CON-15*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The *decision to act* is urgent, even if the first action is only gathering information. Urgency never overrides safety or methodical planning: initial objectives avoid unnecessary risk and unproductive tasking.

The Coordinating Authority usually sets initial urgency shortly after notification. The IMT can review and formalise it during the initial response.

In short: decide urgently, then plan a response that fits the circumstances.

Condensed from draft v0.1, Part 2, principle 9.

2.10 PRN-10 Structured information management

PRN-10 · Principle · version 1

- **Supports:** *CON-11*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Built on the *COP*, information management rests on three tests:

- **Access**: everyone who needs information can see it.
- **Accuracy**: intelligence is grounded in verified evidence; displaying it visually helps catch errors.
- **Actionability**: information is analysed and presented clearly enough to guide objectives.

In short: I need to see what you see. Record what you are doing and why.

Condensed from draft v0.1, Part 2, principle 10.

2.11 PRN-11 Systematic and methodical approach: POA × POD = POS

PRN-11 · Principle · version 1

- **Supports:** *CON-07*, *CON-08*, *CON-12*, *CON-13*, *CON-14*, *CON-15*, *CON-16*, *CON-17*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Operations aim to maximise *POS*. In the initial response this generally follows the objectives of the Bike Wheel model: actions that statistically give the best chance of success. If the subject is not found, the operation moves to extended search planning, where areas with high *POA* are searched in priority order with resources that give the highest *POD*.

Focus on what can be controlled: intelligence gathering, a detailed subject profile, local knowledge and lost person behaviour data, and plausible scenarios that direct the search. Record every plan, action and decision. Proven processes support, but never replace, the IMT's judgement.

In short: focus on what you can control, trust the process, and work in small, well-considered steps.

Condensed from draft v0.1, Part 2, principle 11.

2.12 PRN-12 The principle of simplicity

PRN-12 · Principle · version 1

- **Supports:** *CON-15*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Processes must be simple to understand and easy to apply in real time. Operational tempo and experience vary widely across the country, so any complication should be refined into a simple process that keeps the underlying concept.

In short: keep it simple. It removes ambiguity and builds confidence.

Condensed from draft v0.1, Part 2, principle 12.

2.13 PRN-13 Prioritising safety

PRN-13 · Principle · version 2

- **Supports:** *CON-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Some risk may be accepted to save a saveable life, but the safety of LandSAR personnel is non-negotiable. A robust safety culture includes dynamic risk assessment, fatigue management, thorough briefings, communications discipline and medical care for teams and the subject. Volunteers are only deployed on tasks they are trained and qualified for.

“Our people get to come home.” “We are in the business of saving lives, not trading them.”

In short: safety is an IMT obligation, not a word. It’s their emergency, but our job.

2.13.1 Version history

- **v2** (illustrative): added fatigue management and communications discipline. *CON-02 Risk management and safety* was written against v1, so it is listed as needing review.
- **v1**: first version.

Condensed from draft v0.1, Part 2, principle 13.

2.14 PRN-14 Continuous improvement

PRN-14 · Principle · version 1

- **Supports:** *CON-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The doctrine runs on a feedback loop. Every operation and *SAREX* is debriefed for lessons and actionable insights. These improve local performance, then go to a national subject-matter-expert review point, and on to Police and LandSAR IMT leads for ratification and a doctrine update.

The loop is driven by the sector stakeholders who live with the operational consequences.

In short: improve nationally, and let others learn from our mistakes and achievements.

Condensed from draft v0.1, Part 2, principle 14.

2.15 PRN-15 Iwi and community partnerships

PRN-15 · Principle · version 1

- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Operations respond to community needs and include iwi and Māori involvement where whenua, whānau, tikanga, local knowledge or consequences require it.

Note

The draft has no principle statement for this item yet, and no concept derives from it, so the *Traceability status* page shows it as a coverage gap.

Condensed from draft v0.1, Part 2, principle 15.

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CHAPTER

THREE

PART 3 CORE CONCEPTS AND FRAMEWORK

The "how" of land search management, from fundamental concepts down to guidelines, processes, roles and capability.

3.1 3.1 Fundamental concepts

The central ideas, theories and approaches at the core of the doctrine. Numbering follows the draft. The 3.1 order, and how content is split between 3.1 and 3.2, are still under review.

3.1.1 CON-01 Hierarchical framework of the doctrine

CON-01 · Concept · version 1

- **Derives from:** *PRN-01* v1; *PRN-04* v1; *PRN-14* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The doctrine cascades from enduring principles to specific operational functions:

Level	What it is
Principles	Foundational beliefs and intent behind the approach
Fundamental and core operational concepts	The central theories; the "how"
Guidelines	Recommended practice for a typical operation; not mandatory
Processes and procedures	Established sequences of action and SOPs
Systems	The frameworks that enable processes
Functions	Individual tasks within processes and systems

The whole runs as a **double-loop learning system**: lessons found at the functional level flow upward and can change processes, guidelines, concepts and even principles. This site implements that loop. See *How this doctrine is maintained*.

Proposed national IMT framework

A national framework would professionalise IMTs, give a career pathway and support succession:

- **IMT development pathway**: training and experience required at each level of responsibility.
- **National tutor pool**: tutors drawn from experienced, current IMT practitioners.
- **Regional representatives**: the subject-matter-expert collaboration point for consistent practice.
- **National IMT coordinator**: gives every group a voice and turns operational lessons into doctrine changes.

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Note

The level names in the draft text, its figure and the doctrine brief don't yet match. This site uses principle → concept → guideline → process → form.

Condensed from draft v0.1, Part 3.1 §1.

3.1.2 CON-02 Risk management and safety

CON-02 · Concept · version 1

- **Derives from:** *PRN-13* v1 (**review needed:** PRN-13 is now v2)
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Safety is governed by the LandSAR Safety Management System (*SMS*), which covers training, exercises and operations, and meets both health and safety law and the memorandum of understanding with the Coordinating Authorities. Following it is mandatory for every member.

Safety management starts at the beginning of every operation and continues until everyone is back to normal business. Briefings happen at three levels: the incident controller or safety officer (strategic hazards and reporting), operations (task-specific hazards) and team leaders (how the team will manage risk).

IMT obligations

1. Cover safety in every team briefing.
2. Assess and record operation-level hazards.
3. Prepare a safety plan, and use it to populate the hazard section of each tasking.
4. Monitor progress, update teams on changing hazards and look after their wellbeing.
5. Debrief teams on safety issues.
6. Record and report accidents and incidents.
7. Stay current with safety advisories and the LandSAR safety standards.

Teams manage dynamic risk in the field with the Take 5 process. Groups benefit from an annual hazard register for their area, so briefings can focus on task-specific risks.

Risk domain	Example controls
Operational environment	Weather and river thresholds, daylight limits, go/no-go decisions
Fatigue	Shift planning, rest periods, replacement teams, IMT fatigue monitoring
Comms and tracking	Scheduled calls, check-in/out, team tracking, lost team procedure
Psychological wellbeing	Pre-briefs for traumatic exposure, peer and post-incident support
Legal and ethical	Privacy, media discipline, scene preservation, safeguarding
Volunteer sustainability	Reasonable call-out expectations, employer support, avoiding burnout

Condensed from draft v0.1, Part 3.1 §2.

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3.1.3 CON-03 Pre-operational readiness plans

CON-03 · Concept · version 1

- **Derives from:** *PRN-05* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

IMTs develop and practise pre-plans for predictable events and locations, based on the area's SAR history. Pre-plans reduce the research and planning burden when an urgent response is needed. They can also cover how the IMT will manage civil defence responses or large numbers of spontaneous volunteers.

A pre-plan typically covers:

- activation and mobilisation
- the area, its geography, access points and typical activities
- initial actions and resources for common scenarios
- health and safety, and environmental conditions
- previous searches in the area
- communications plan and coverage
- key contacts, facilities, medical and logistics resources.

Keep pre-plans brief, and structure them consistently so people can find information quickly.

Condensed from draft v0.1, Part 3.1 §3.

3.1.4 CON-04 Notification and awareness

CON-04 · Concept · version 1

- **Derives from:** *PRN-02* v1; *PRN-09* v1
- **Supports:** *PRC-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Notification starts every operation. Reports usually reach a Coordinating Authority directly or through partners such as ambulance services or the Department of Conservation. All reports should be channelled to the Coordinating Authority, so urgency and response can be assessed with local knowledge.

Whoever receives and assesses the report becomes the **incident controller** and holds all *CIMS* functions until they delegate them. Their first jobs are to record the key information (the Missing Person Report, *MPR*, is a good guide) and to assess the likely scale and urgency.

- **Urgent and obvious**: deploy immediately using local knowledge and available assets.
- **Not obvious**: work through the first steps of the *initial response process* (MPR, map, information collection, obvious scenarios) before finalising search urgency, because urgency scored on thin information tends to overstate it.

Avoid planning alone. A LandSAR IMT can be engaged early to support these decisions, and should always set up as if the operation will be protracted.

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Note

Whether notification is in scope for this doctrine is still to be confirmed against the doctrine brief.

Condensed from draft v0.1, Part 3.1 §4.

3.1.5 CON-05 Leadership

CON-05 · Concept · version 1

- **Derives from:** *PRN-06* v1; *PRN-03* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

LandSAR works as a support agency to the Coordinating Authorities, who hold ultimate authority. Within that relationship, leadership is a cornerstone of a high-performing IMT, both within the team and between teams and agencies.

Strong leaders collaborate, communicate clearly, maintain situational awareness and seek expertise when needed. Leadership is developed through training, experience on the job and succession planning.

Condensed from draft v0.1, Part 3.1 §5.

3.1.6 CON-06 Incident management teams and CIMS

CON-06 · Concept · version 1

- **Derives from:** *PRN-02* v1; *PRN-03* v1
- **Supports:** *GDL-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

An IMT plans and manages the whole operation so it is safe, efficient and effective. An IMT should be set up for every operation, whatever its scale, so no one plans alone. Structure and formality scale with the operation: be practical at the start, then formalise roles as complexity and span of control grow.

CIMS function	Role in a land search
Control	Overall strategy, objectives, safety and coordination; appointed by the lead agency. A response manager can run the IMT day to day.
Intelligence	What we know, what we need to know; subject profile, planning points, weather, terrain, field intelligence; builds the *COP*
Planning	Where to search next; *IAP*s, long-term and contingency plans
Operations	Tasking, coordinating and supervising field teams and air assets
Logistics	People, equipment, transport, food, facilities; responder wellbeing before and after tasks
Safety	Hazard identification and a safe work culture across all agencies
Welfare	Support for the subject's family and the affected community
Public information	Media and community messaging, usually with Police

Getting search teams to the right place

- Know the capability of local and neighbouring resources, and engage them early.
- Seek expert advice for unfamiliar tasks or resources.
- Review each operational period to check intelligence has been acted on.
- Use limited resources early to gather information.
- Use trained IMT members, and assume a protracted operation from the start.
- Work with local communities.

Diverse teams, with field, management and analytical backgrounds, make better decisions and avoid groupthink.

Condensed from draft v0.1, Part 3.1 §6.

3.1.7 CON-07 Phases of a land search and rescue operation

CON-07 · Concept · version 1

- **Derives from:** *PRN-11* v1
- **Supports:** *GDL-04*, *GDL-06*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

An operation is a continuum, but each phase needs different thinking:

1. **Initial response (*IR*)**: pre-determined actions based on lost person behaviour and the *Bike Wheel model*; CIMS functions applied informally and scaled to need.
2. **Extended search planning (*ESP*)**: starts if the initial response doesn't find the subject. Planning shifts from the statistical subject to *this* subject, using scenarios, consolidated intelligence and methodical searching of probable areas.
3. **Rescue (known location)**: can start at any point when the subject is found, or as a stand-alone operation, for example a beacon activation.

Notification isn't a phase as such; it usually sits with a Coordinating Authority.

Intelligence is the fuel. Without enough focus on collecting and analysing it, a search gets stuck in the initial response, tasking resources reflexively with no evidence to prefer one place over another. Move to extended search planning as the initial response objectives are completed, and resource intelligence from the start so the transition is smooth.

Note

The draft uses several names for the second phase and differs on whether "operational conclusion" is a fourth phase. This site uses *extended search planning* throughout.

Condensed from draft v0.1, Part 3.1 §7.

3.1.8 CON-08 Maths in SAR

CON-08 · Concept · version 1

- **Derives from:** *PRN-08* v1; *PRN-11* v1

- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Numbers run through land SAR: search area and travel distance, search urgency scores, lost person behaviour statistics, scenario weighting and probability of area. People interpret probability *words* very differently; numbers remove that ambiguity, even when the judgement behind them is subjective.

POD is often reported as High, Medium or Low. Words alone can't be used to recalculate *POA* after a segment is searched, which is central to progressive search management. The doctrine therefore binds each word to a number. A simple prompt: "If there were 10 people in your area, how many would you have seen?" Eight means 80 %.

Rating	POD as drafted
High	66–99 %
Medium	33–66 %
Low	1–33 %

This scale is defined once, in `_snippets/pod-scale.md`, and included on every page that shows it. The draft's band boundaries overlap at 33 and 66; fixing them here fixes every page.

Recalculating probability is covered in *CON-17 Search theory: POA × POD = POS*.

Condensed from draft v0.1, Part 3.1 §8.

3.1.9 CON-09 Lost and missing person behaviour

CON-09 · Concept · version 1

- **Derives from:** *PRN-08* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Lost person behaviour (*LPB*) research, built from large international incident databases, moves initial planning from instinct to data. It lets an IMT set objectives and send resources to the most likely places first, when little is known about the subject. It is a tool, not the answer.

Apply it with judgement:

- The data is aggregated. Specific knowledge of this subject (fitness, injury, state of mind, intentions) can outweigh it.
- Local SAR history matters: subjects cluster where terrain and common decision errors take them.
- Check the **ecoregion** and the **sample size** (n) behind each figure; small samples are less reliable.
- Choose the **subject category** (for example hiker, child, despondent, dementia) using the category hierarchy; several may need considering.
- Use the category's **investigative questions** and **initial tasks**.
- Apply the category tables to the search area: distance travelled, elevation change, mobility, dispersion, find location, scenarios, track offset and survivability.

LandSAR runs dedicated training on applying LPB.

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Note

Figures for New Zealand's share of the data, and success rates attributed to applying the model, are being verified and are not reproduced here.

Condensed from draft v0.1, Part 3.1 §9.

3.1.10 CON-10 Communications

CON-10 · Concept · version 1

- **Derives from:** *PRN-07* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Communications connect every level, from the subject to the incident controller. The IMT is the hub: it receives distress signals, status reports and clues, turns them into intelligence, coordinates internally, and sends a clear action plan back out. That feedback loop validates the strategy and lets it adjust.

The IMT needs clear, concise:

- briefings in a standard format, with safety included
- taskings with scheduled reporting times
- status reports and structured IMT meetings
- handovers to incoming IMTs
- radio or satellite communication with field teams
- planned debriefs of every team.

Field teams provide radio checks, situation reports on schedule, early intelligence on clues and conditions, and a POD from each searcher after the task.

Capability	Doctrine position
Radio	Interoperable channels, disciplined procedure, fallbacks for terrain shadow
Mapping and GIS	One common picture of tasks, tracks, clues, hazards and probability areas
Drones and aircraft	Use where lawful, safe and relevant; don't overclaim effectiveness in dense bush
Mobile and digital data	Use Police-approved processes; treat digital clues as perishable and private
Data protection	Role-based access, minimal disclosure, secure storage

Prefer simple, robust systems that work when infrastructure is degraded.

Condensed from draft v0.1, Part 3.1 §10.

3.1.11 CON-11 Evidence-based and intelligence-led operations

CON-11 · Concept · version 1

- **Derives from:** *PRN-08* v1; *PRN-10* v1
- **Supports:** *GDL-01*

- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

- **Evidence-based**: choose strategies by appraising proven data (historical SAR statistics, research, professional expertise) rather than habit.
- **Intelligence-led**: continuously collect, process and analyse information during the operation, and use it to set priorities, adjust resources and change tactics.

Early in the initial response, statistics drive most actions. As the operation progresses, intelligence makes the strategy specific to this subject and this terrain. Record every plan, action and decision throughout.

The intelligence function turns information into intelligence under three tests: access, accuracy and actionability. Source reliability can be graded where people are trained to do so.

Common operating picture

The *COP* is the single, shared display of the incident: everyone works from the same verified facts, decisions are faster, and the IMT can see coverage and team positions at a glance. A SAR COP typically shows:

1. timeline (weather, subject actions, SAR actions)
2. current incident action plan
3. subject profile
4. information collection plan
5. weather and terrain overlays
6. planning map (planning points, hazards, attractants, clues, regions and segments, statistical rings, search effort)
7. operations map (team positions and status)
8. planning data (urgency, LPB category, scenarios, clues)
9. logistics and resource tracking.

Condensed from draft v0.1, Part 3.1 §11.

3.1.12 CON-12 Management by objectives

CON-12 · Concept · version 1

- **Derives from:** *PRN-11* v1
- **Supports:** *GDL-01*, *GDL-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Management by objectives (*MBO*) keeps every action tied to the mission, through short, time-bound operational periods. It is consistent with CIMS. Teams that jump straight to tasks usually have objectives in mind; writing them down creates clarity.

Remember ***MOST***:

	Question
Mission	Why: for example “locate the missing person”
Objectives	What will we do this period?
Strategies	How will we do it?
Tasks	Who will do it?

Objectives should be **SMART**: specific, measurable, achievable, relevant and time-bound.

Short operational periods (often 8–12 hours, aligned with Coordinating Authority shifts) allow the plan to adapt to new information, manage fatigue, make progress measurable, and bring fresh thinking as IMTs rotate.

Condensed from draft v0.1, Part 3.1 §12.

3.1.13 CON-13 Incident action plans

CON-13 · Concept · version 1

- **Derives from:** *PRN-11* v1; *PRN-05* v1
- **Supports:** *GDL-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The incident action plan (*IAP*) puts MBO into practice: a concise, forward-looking plan for one operational period that balances probability, detection, survivability, hazards, urgency and the consequences of being wrong. The planning function leads it, with input from every function.

- **Urgent searches**: deploy first and back-capture the IAP. Uncertainty is not a reason for delay.
- **Protracted searches**: write the IAP before teams are briefed and deployed.
- **Handovers**: never hand over without an IAP for the next period.

Building an IAP

1. **Set the operational period first**, allowing for travel and debrief. For example, a 10-hour period might be 2 hours travel, 4–6 hours searching, 2 hours return and debrief. Searching much longer than 4–6 hours degrades POD.
2. **Set prioritised SMART objectives**, typically covering search, safety, management and investigation.
3. **Develop the plan**: objectives (what), strategies (how), tasks (who).
4. **Task only to objectives.** If a new task emerges, change the plan formally rather than tasking ad hoc.
5. **Debrief and feed back** into the next IAP.

Use a simple one-page IAP per operational period, filed with that period’s tasks and debriefs, so the record builds period by period.

Condensed from draft v0.1, Part 3.1 §13.

3.1.14 CON-14 The Bike Wheel model

CON-14 · Concept · version 1

- **Derives from:** *PRN-11* v1
- **Supports:** *GDL-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The Bike Wheel is a visual model that turns lost person behaviour statistics into initial response objectives. It comes from international search theory and has been adapted in New Zealand to include management and intelligence objectives.

Wheel element	Initial response objective
Hub	Process the planning point and search close in
Spokes	Search the intended route and travel corridors
Reflectors	Check hazards, attractants and areas with SAR history
Rim, with rocks at corridor ends	Contain the area at natural containment points
Stick in the wheel	Search areas suggested by early scenarios
Frame	Build the profile, establish comms, prepare for handover

Example initial response objectives (one operational period)

1. Complete the information collection plan and build the subject profile.
2. Establish reliable communications for the search area.
3. Contain the search area.
4. Process the planning point and complete a close-in search.
5. Search the area suggested by the initial scenario.
6. Search the intended route and check decision points.
7. Search travel corridors next to the intended route.
8. Check known hazards, attractants and SAR history locations.
9. Prepare the IAP for handover.

In extended search planning the same wheel is reused: what is left is “the spaces between the spokes”, which are reduced, divided into regions and segments, and weighted.

Note

Fixed distances shown in some versions of the figure, and attribution, are still being checked.

Condensed from draft v0.1, Part 3.1 §14.

3.1.15 CON-15 The initial response process

CON-15 · Concept · version 1

- **Derives from:** *PRN-09* v1; *PRN-11* v1; *PRN-12* v1
- **Supports:** *GDL-02*, *PRC-01*

- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The initial response begins straight after notification and follows pre-determined actions guided by the *Bike Wheel*. Its length depends on the size and complexity of the area, not a fixed time.

IMT managers need a process they can repeat without reference material. The mnemonic is ***MMISSI TD***:

1. **M**issing person report: initial details, medical, equipment, intentions.
2. **M**ap: terrain analysis, hazards, LPB search area, travel time and distance, planning points.
3. **I**nformation collection plan: build the profile, fill gaps, assign enquiries.
4. **S**cenario: quick, obvious scenarios.
5. **S**earch urgency: confirm the initial assessment still holds.
6. **I**ncident action plan: set objectives with the Bike Wheel.
7. **T**ask: from the objectives; preserve the planning point and clues; contain.
8. **D**ebrief: gather intelligence and POD.

Repeat steps 6–8 until the initial response objectives are met. The step-by-step version is *PRC-01 Initial response (MMISSI TD)*.

Note

How urgent cases bypass the sequence, and how steps run in parallel, is still being worked through.

Condensed from draft v0.1, Part 3.1 §15.

3.1.16 CON-16 The extended search planning process

CON-16 · Concept · version 1

- **Derives from:** *PRN-11* v1
- **Supports:** *GDL-05*, *PRC-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Extended search planning builds a progressive plan on the results of earlier search effort. In outline:

1. Review the initial response (start while it winds down).
2. Determine the search area by subjective and deductive reasoning.
3. Develop scenarios that fit the evidence, LPB data and profile.
4. Mark scenario and geographical regions on the map.
5. Weight the regions.
6. Divide the highest-weighted regions into segments.
7. Weight the segments.
8. Task to the highest-weighted segments.
9. Debrief teams to establish POD.

10. Recalculate segment weights from POD and new intelligence.
11. Re-task to the next highest regions or segments.

Repeat steps 8–11. Because every segment carries a number, segments in different regions can be compared directly. The step-by-step version is *PRC-02 Extended search planning cycle*.

Condensed from draft v0.1, Part 3.1 §16.

3.1.17 CON-17 Search theory: POA × POD = POS

CON-17 · Concept · version 1

- **Derives from:** *PRN-08* v1; *PRN-11* v1
- **Supports:** *GDL-05*, *GDL-06*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Search planning comes down to two things: finding the right place to search, and applying resources that can detect what you are looking for, at the right time.

$$POS = POA \times POD$$

- *POA*: the likelihood the subject is in this region or segment.
- *POD*: the likelihood this resource would find the subject if they were there.
- *POS*: the overall likelihood of success.

Closed search system

New Zealand uses a **closed search area**: the IMT defines a bounded area and assumes the subject is inside it. The rest of the world is not forgotten; it is covered by investigation and public appeals, not by searching.

Search area, regions and segments

In the initial response the search area is set by LPB statistical rings, theoretical travel distance and any early scenarios. In extended search planning it is refined into a shape that follows terrain and evidence; the rings remain for reference only.

The refined area is divided into about 6–10 **regions** along clear terrain features: scenario-based regions (usually higher POA) and geographical regions. High-POA regions are divided into **segments**, each sized for a 4–6 hour task.

Scenarios and weighting

A scenario is a story that runs from the planning point to a specific piece of terrain. “Lost somewhere in the area” is not useful. Present all the evidence to the IMT first, then test each scenario: does it fit the facts, is it realistic, is it consistent with LPB and the profile, is it achievable in this terrain, and what evidence contradicts it?

Weight scenarios with the *scenario weighting sheet* or, where evidence is complex, with analysis of competing hypotheses (*ACH*), which looks for the scenario *least contradicted* by the evidence. The two are complementary.

Recalculating after a search

When a segment is searched and the subject isn’t found, its POA drops:

$$POA_{new} = POA_{old} \times (1 - POD)$$

For example, a segment weighted 14 searched at 80 % POD drops to 14 × 0.2 = 2.8. A second search with a different resource reduces it further. The IMT, not the field team, owns the final POD: it checks the debrief against GPS tracks, photos and conditions before using it, because an inflated POD moves the search away from a poorly searched area.

Rating	POD as drafted
High	66–99 %
Medium	33–66 %
Low	1–33 %

This scale is defined once, in `_snippets/pod-scale.md`, and included on every page that shows it. The draft's band boundaries overlap at 33 and 66; fixing them here fixes every page.

Note

How ACH results translate into segment weights, and tools for more objective POD, are open questions in the draft.

Condensed from draft v0.1, Part 3.1 §17.

3.2 3.2 Core operational concepts

The concepts from 3.1, applied in an operational context. The draft lists the topics; content is still to be written.

Not yet drafted

Each topic will become an item page that derives from the 3.1 concept it applies.

1. IMT set-up and the common operating picture
2. CIMS in practice
3. Lost person behaviour
4. Information management and intelligence
5. Search urgency
6. Planning and maps
7. Planning points
8. Scenario development
9. Search area determination in IR and ESP
10. Close-in search
11. Containment
12. Decision points
13. Terrain analysis
14. IAPs and the Bike Wheel
15. Search methods
16. Tasks, briefing and the mobility matrix

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17. Search POD and coverage (search effort evaluation)
18. Rescue
19. Team debriefing
20. Handovers
21. Conclusion: demobilisation, suspension, welfare, actionable insights and operational debriefs

3.3 3.3 Guidelines

Guidelines are a consolidated overview of the recommended process for a typical land search. They are not mandatory, but they support consistency and effective practice. Pick the section relevant to you on the day.

Guidance tier: one approval from any reviewer merges a change.

3.3.1 GDL-01 Managing an IMT: key actions

GDL-01 · Guideline · version 1

- **Derives from:** *CON-06* v1; *CON-11* v1; *CON-12* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Carry these out continuously, from notification to conclusion, in every operation:

- **Develop the subject profile** through ongoing interviews and investigation.
- **Manage risk dynamically** until everyone is home after the final debrief.
- **Scale the structure**: apply *CIMS* in proportion to the operation.
- **Prioritise the four Cs** in the initial response: confirm the planning point, close-in search, containment, communications. Immediate hazards or a strong scenario can take priority.
- **Document everything**: intelligence, decisions, resources, actions, debriefs.
- **Keep a common operating picture** on walls or shared displays, not hidden on individual screens.
- **Plan for a long operation** from the outset.
- **Engage help early**: catching up once behind is very hard.
- **Treat search teams as intelligence sources**: look for clues and sign, not just the subject.
- **Manage media and social channels** with a dedicated resource, usually through Police.
- **Use LPB data and the Bike Wheel** in the initial response.
- **Process before forms**: keep the method going even when forms or technology are unavailable.
- **Manage by objectives**: don't confuse being busy with being effective.

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Setting up

Topic	Guideline
Notification	The Coordinating Authority is incident controller until it delegates. Use the MPR. If urgent, deploy now; otherwise put teams on standby and follow the initial response process. Avoid planning alone.
Safety	Develop a safety plan early. Appoint a safety officer if the environment is beyond the controller's expertise. Only send trained and equipped people.
Pre-plans	Check for and action any relevant pre-plan.
Incident control point	Choose a location with access, power, comms, display space and room to scale. Staff the IMT so no function is compromised.
Forward base	Decide its purpose first. Watch for isolated functions, loss of control by the IMT and intelligence that doesn't get passed on.

Condensed from draft v0.1, Part 3.3.1.

3.3.2 GDL-02 Initial response phase

GDL-02 · Guideline · version 1

- **Derives from:** *CON-15* v1; *CON-14* v1
- **Supports:** *FRM-03*, *PRC-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Follows the *initial response process*. Start an operations log, decision log and timeline straight away, and display everything on the common operating picture.

Step	Guideline
1. Missing person report	Complete or update the MPR. Build the profile and choose an LPB category.
2. Map	Mark planning points, close-in search area, LPB rings and intended route. Analyse travel corridors, hazards, weather effects on terrain and subject, barriers and attractants.
3. Information collection plan	Address the gaps from steps 1 and 2. Assign enquiries, record results, display key intelligence.
4. Initial scenarios	Quick and obvious stories that point to a piece of terrain: start point, direction of travel, route, activity, destination. Rank by consensus.
5. Search urgency	Confirm or update the Coordinating Authority's assessment. Consider go/no-go for responder risk: don't make their emergency your emergency.
6. Incident action plan	Set the operational period and mission, consider the factors, write SMART objectives, strategies and tasks.
7. Tasking	Specific, unambiguous tasks of 4–6 hours. State whether speed, efficiency or thoroughness matters most (*SET*). Brief teams together in a standard format.
8. Debrief	The most critical step. Debrief the whole team, not just the leader. Record POD from each searcher. Feed results into the next IAP.

Recording POD at the debrief

Ask each searcher for a rating, then confirm it as a number. Average the individual values; outliers show where re-searching is needed.

Rating	POD as drafted
High	66–99 %
Medium	33–66 %
Low	1–33 %

This scale is defined once, in `_snippets/pod-scale.md`, and included on every page that shows it. The draft's band boundaries overlap at 33 and 66; fixing them here fixes every page.

Condensed from draft v0.1, Part 3.3.2.

3.3.3 GDL-03 Handovers, demobilisation and operational debriefs

GDL-03 · Guideline · version 1

- **Derives from:** *CON-13* v1; *CON-12* v1
- **Supports:** *FRM-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Note

This section is empty in the draft. The points below come from the draft's summary checklist.

Handover

- Prepare the briefing area and update maps.
- Complete the operational period summary.
- The incoming IMT arrives early; the outgoing IMT briefs it using the IAP structure.
- The incident controller stays on briefly after handover.

Demobilisation

- Set release priorities and plan safe extraction of teams.
- Keep communications until everyone is accounted for.
- Return equipment and complete sign-out.
- Look after welfare: food, warmth, rest, critical incident stress.
- Hold a hot debrief and schedule a formal debrief.

Condensed from draft v0.1, Part 3.3.3.

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3.3.4 GDL-04 Rescue phase: known location

GDL-04 · Guideline · version 1

- **Derives from:** *CON-07* v1
- **Owner:** doctrine-group · **Last reviewed:** 2025-06-30

Rescue can start at any point, or as a stand-alone operation such as a beacon activation where the nature of the emergency is unknown until the party is reached. Finding the subject changes the mission but doesn't end the operation.

1. **Locate**: confirm position precisely, assess hazards, notify the IMT, protect the scene, prevent unnecessary convergence.
2. **Stabilise**: first aid, protection from the environment, reassurance, ongoing monitoring.
3. **Plan extraction**: walk-out, stretcher, rope, helicopter, boat or vehicle, matched to the subject's condition, weather, daylight, terrain and responder fatigue.
4. **Transfer care** with a structured medical handover, recording times, interventions and condition changes.
5. **Family and media** go through Police and the IMT.

If a fatality or crime is possible, preserve evidence, avoid unnecessary movement, record observations and follow Police direction.

Condensed from draft v0.1, Part 3.3.4.

3.3.5 GDL-05 Extended search planning phase

GDL-05 · Guideline · version 1

- **Derives from:** *CON-16* v1; *CON-17* v1
- **Supports:** *FRM-03*, *PRC-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Move to extended search planning as the initial response objectives are completed; start planning before they finish so the transition is seamless.

Step	Guideline
Review the IR	Map areas searched, clues and POD. Check paperwork. Identify gaps in intelligence and coverage, and whether re-searching with other techniques would help.
Search area	Move from circles to shapes: bound the area by bluffs, rivers, ridges and corridors, using the subject's profile, capability, equipment and clues.
Scenarios	Present all intelligence to the IMT first. Test each scenario against the facts, terrain and LPB. Record each on a scenario recording sheet and map it as a region.
Regions	Aim for about 6–10, mixing scenario and geographical regions.
Weighting	Use the *scenario weighting sheet* (option A) or ACH (option B). Write the weights on the map.
Segments	Segment the top one to three regions; weight segments using the same scenario.
Task and debrief	Task the highest segments; debrief whole teams for POD.
Recalculate	Reduce segment weights by POD. Re-search with a different resource, direction or timing, or move on.

You will never reach 100 %. Deciding when to move on is a judgement informed by the numbers and the quality of the POD.

Condensed from draft v0.1, Part 3.3.5.

3.3.6 GDL-06 Suspension

GDL-06 · Guideline · version 1

- **Derives from:** *CON-07* v1; *CON-17* v1
- **Supports:** *FRM-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Note

This section is under review with the Coordinating Authority.

Suspension is a Coordinating Authority decision. LandSAR members provide input and advice but don't make it.

Options are a **limited continuous search** (flyers, media), **scaling down** to fewer assets in a smaller area, or **suspending** until new information comes to light.

An IMT preparing suspension evidence provides:

- a summary of all search actions
- a map of areas searched, coverage and gaps
- updated POA and POD analysis
- clues found and their relevance
- scenario analysis and weighting outcomes
- a specialist survivability assessment
- a risk assessment for continued searching
- a record of family liaison and the family's wishes.

The operation is then peer reviewed by someone not involved, the family is told, and approval is sought from the appropriate authority for the operation's category.

Condensed from draft v0.1, Part 3.3.6.

The summary checklist (3.3.7) is a form: *FRM-03 IMT summary checklist*.

3.4 3.4 Processes and procedures

The established sequences of action that put the guidelines into practice. Each process derives from the concepts and guidelines it implements, so a change to any of them flags the process for review.

A land search runs through four stages: initial response, extended search planning, rescue, and operational conclusion (demobilisation, handovers, debriefs and, if needed, suspension).

3.4.1 PRC-01 Initial response (MMISSI TD)

PRC-01 · Process · version 1

- **Derives from:** *CON-15* v1; *GDL-02* v1
- **Supports:** *FRM-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Step	Do	Output
1	Complete or update the missing person report	MPR, LPB category
2	Map and terrain analysis	Planning map with planning points, rings, route, hazards
3	Information collection plan	Assigned enquiries, updated profile
4	Initial scenarios	Ranked scenarios tied to terrain
5	Confirm search urgency	Search urgency form, go/no-go
6	Incident action plan	IAP with Bike Wheel objectives
7	Task	Tasking sheets, briefing
8	Debrief	*Team debrief* with POD

Repeat steps 6–8 until the initial response objectives are met, then start *PRC-02 Extended search planning cycle*.

Condensed from draft v0.1, Parts 3.1 §15 and 3.4.

3.4.2 PRC-02 Extended search planning cycle

PRC-02 · Process · version 1

- **Derives from:** *CON-16* v1; *GDL-05* v1
- **Supports:** *FRM-01*, *FRM-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Step	Do	Output
1	Review the initial response	Map of effort, clues, gaps
2	Determine the search area	Refined search area boundary
3	Develop scenarios	Scenario recording sheets
4	Mark regions	6–10 scenario and geographical regions
5	Weight regions	*Scenario weighting sheet*
6	Segment top regions	Segments sized for 4–6 hour tasks
7	Weight segments	Segment POA
8	Task highest segments	Tasking sheets
9	Debrief	*Team debrief* with POD
10	Recalculate	Updated segment POA
11	Re-task	Next IAP

Repeat steps 8–11 until the high-POA segments are searched to the IMT's satisfaction.

Condensed from draft v0.1, Parts 3.1 §16 and 3.4.

3.4.3 PRC-03 Notification and on-scene coordination

PRC-03 · Process · version 1

- **Derives from:** *CON-04* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

1. Reports go to the Coordinating Authority, which becomes incident controller.
2. The controller assesses scale, urgency and likely resources, ideally with an IMT rather than alone.
3. The Coordinating Authority engages LandSAR for field teams, an IMT, or both.
4. On-scene coordination:
    - If RCCNZ engages LandSAR without Police, a LandSAR response manager is assigned as on-scene coordinator (*OSC*).
    - If Police are involved and a LandSAR IMT is set up, Police can take the OSC role.

Note

The on-scene coordinator arrangements are being verified with the Coordinating Authorities.

Condensed from draft v0.1, Part 3.4.

3.4.4 Standard operating procedures

National SOPs or checklists proposed in the draft:

- helicopter briefing
- minimum field team and safety equipment
- “actions on” contingencies.

Groups may add local SOPs, for example: standing up the IMT, forward base management, spontaneous volunteer management, out-of-area deployment, search dog and drone deployment, and night operations.

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3.5 3.5 Roles and responsibilities

Who does what in relation to the doctrine.

Actor	Role	Responsibilities for the doctrine
NZSAR Council	National strategic governance of the SAR system	Align sector guidelines with the doctrine; feed sector lessons into it
New Zealand Police	Coordinating Authority for Category I operations, including most land SAR	Endorse the doctrine; deploy LandSAR often to build capability; contribute to processes and debriefs
RCCNZ	Coordinating Authority for Category II operations; directs a growing number of beacon-initiated land operations, usually through an on-scene coordinator	Endorse the doctrine; integrate land teams where needed
Police SAR squads	Local knowledge, subject information, joint training with groups	Contribute at all levels, especially processes and debriefs
LandSAR national office	Custodian of the doctrine; supports the federation of volunteer groups	Maintain the doctrine; align training, systems and readiness reporting
Local LandSAR groups	Volunteer response capability and local knowledge	Train in and apply the doctrine; maintain readiness to agreed standards
Iwi, whānau, community and land managers	Local knowledge, tikanga, access, family connection	Engage early and respectfully, especially where whenua, marae, wāhi tapu or private land are involved

The doctrine is freely available to sector partners, and every organisation with relevant expertise is invited to contribute through the continuous improvement process.

Note

IMT function roles (control, intelligence, planning, operations, logistics, safety, welfare, public information) are currently described in *CON-06 Incident management teams and CIMS*. Where they should sit is still under review.

Condensed from draft v0.1, Part 3.5.

3.6 3.6 Capability and resources

The doctrine supports existing national doctrine such as *CIMS*. To deliver consistent practice, all sector partners need to agree on it as the baseline, then refine it over time.

Beyond process, the draft identifies four enablers that need sector-wide commitment:

1. consistent national call-out and engagement procedures
2. an interoperable mapping tool
3. an interoperable incident management system
4. communication protocols, processes and resources.

Together these support activation and a common operating picture at local, regional and national level, and should extend to working alongside civil defence.

Condensed from draft v0.1, Part 3.6.

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CHAPTER

FOUR

PLANNED PARTS

Later parts are not in this review round. Topics known so far are listed so overlaps with Parts 2–3 can be resolved before drafting.

Planned section	Overlaps with
Part 4 Application and implementation	
4.1 Planning and preparation	*CON-03 Pre-operational readiness plans*
4.3 Leadership	*PRN-06 Effective and supportive leadership*, *CON-05 Leadership*
4.4 Communication	*PRN-07 The critical role of communications*, *CON-10 Communications*
4.5 Training	National IMT framework in *CON-01 Hierarchical framework of the doctrine*
Part 5 Review and evolution	
5.2 Lessons learned	*PRN-14 Continuous improvement*

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CHAPTER

FIVE

FORMS

Forms derive from processes or guidelines, so a change upstream flags the forms that need checking.

5.1 FRM-01 Team debrief (search effort evaluation)

FRM-01 · Form · version 1

- **Derives from:** *PRC-01* v1; *PRC-02* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Debrief the whole team in person where possible: one person asks (operations), one records (intelligence). Don't let teams fill this in themselves.

Field	Notes
Team, task and segment	
Time in and out	
Route and coverage	Annotated map and GPS track
Technique and spacing	As tasked, or what changed and why
Conditions	Vegetation, visibility, weather, terrain
Clues	What, where, when; photographed and preserved?
Hazards found	Pass to the safety function
POD per searcher	Rating and number; average them
What limited detection	Fatigue, distractions, radio issues, terrain
Recommendations	Re-search, other resources, new areas

Rating	POD as drafted
High	66–99 %
Medium	33–66 %
Low	1–33 %

This scale is defined once, in `_snippets/pod-scale.md`, and included on every page that shows it. The draft's band boundaries overlap at 33 and 66; fixing them here fixes every page.

Fields drawn from draft v0.1, Parts 3.1 and 3.3.

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5.2 FRM-02 Scenario weighting sheet

FRM-02 · Form · version 1

- **Derives from:** *PRC-02* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Use after the evidence has been presented to the whole IMT.

1. Each person independently gives their most likely scenario a score of 100.
2. They score every other scenario from 10 to 100, relative to it: 50 means half as likely. More than one scenario can score 100.
3. Add each scenario's scores.
4. Divide each total by the sum of all totals and multiply by 100 to get the weighting.

5.2.1 Worked example

Scenario	Member 1	Member 2	Member 3	Total	Weighting
A Injured near main track	100	80	100	280	40
B Side trip to a lake	80	100	70	250	36
C Lost in bush off track	30	40	20	90	13
D Sheltering off track	20	30	30	80	11
Sum				**700**	**100**

The weightings are relative values for setting priorities, written on the map as each region's POA.

Method from draft v0.1, Part 3.1 §17. Example values are invented.

5.3 FRM-03 IMT summary checklist

FRM-03 · Form · version 1

- **Derives from:** *GDL-02* v1; *GDL-03* v1; *GDL-05* v1; *GDL-06* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

A rapid-reference record for people who know the process.

5.3.1 Notification and safety

- Informant details recorded; MPR started
- Collaborating early, not planning alone
- Response decision made; teams on standby if not urgent
- Operations log, decision log and timeline started
- Safety plan started; safety officer appointed if needed

5.3.2 Incident control point and information

- Location chosen (power, comms, space, scalable)
- Common operating picture set up
- IMT roles assigned, including family liaison and media
- Profile, information collection plan and enquiries under way

5.3.3 Map and initial scenarios

- Planning points, close-in search area and route plotted
- LPB rings and theoretical distance plotted
- Hazards, attractants, barriers and corridors marked
- Quick scenarios ranked; most plausible set as an objective

5.3.4 Urgency, IAP and tasking

- Search urgency confirmed; go/no-go for responder risk
- Operational period and mission set; SMART objectives, strategies and tasks
- Tasks of 4–6 hours with map, subject summary and hazards
- Teams briefed together

5.3.5 Debrief

- Whole team debriefed; GPS, photos and maps reviewed
- POD recorded per searcher; clues analysed
- Results fed into the next IAP; POA updated

5.3.6 Extended search planning

- IR reviewed; search area refined
- Regions and segments defined and weighted
- Intelligence gathering continues

5.3.7 Handover, demobilisation and suspension

- Operational period summary complete; incoming IMT briefed
- Teams extracted and accounted for; welfare managed
- Hot debrief held; formal debrief scheduled
- If recommending suspension: evidence pack prepared (see *GDL-06 Suspension*)

Condensed from draft v0.1, Part 3.3.7.

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CHAPTER

SIX

FIGURES

Diagrams of four ideas from Part 3, drawn for this demonstration from the condensed text, and two logos as examples of raster images. The diagrams aren't the draft's own figures.

Note

Diagrams are SVG files next to this page, so changes to them are reviewed in a pull request like text; the PDF build converts them. Photos and logos are PNG or JPG files, used as they are. Figures are numbered automatically and can be cross-referenced: the search planning loop is in Figure 6.4.

6.1 Levels of the doctrine

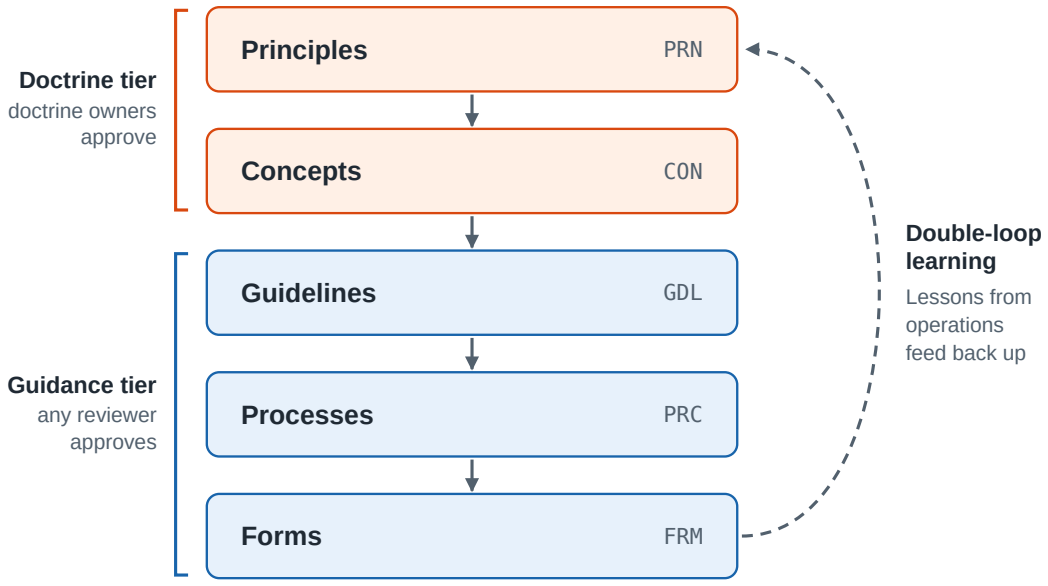


Figure 6.1: The levels this site uses, each deriving from the one above, and the review tier for each (*CON-01*). Lessons from operations flow back up and can change any level.

6.2 Phases of an operation

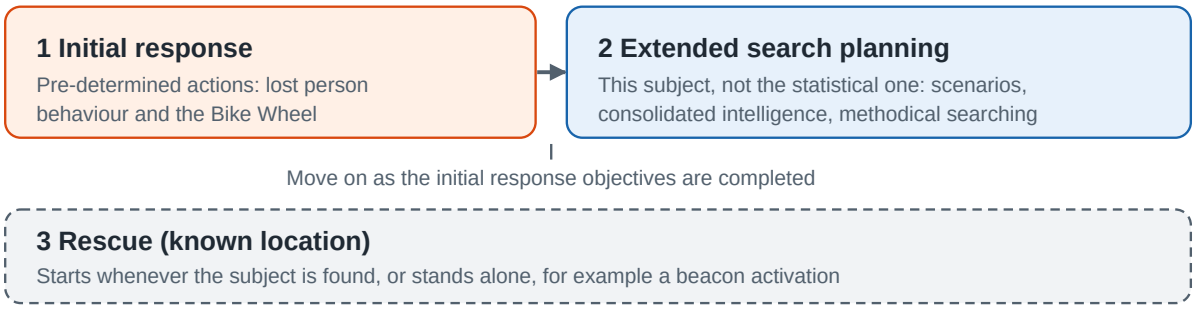


Figure 6.2: The three phases (*CON-07*). Rescue can start whenever the subject is found.

6.3 The Bike Wheel model

6.4 Extended search planning

6.5 Raster images

PNG and JPG files need no conversion, so the HTML and the PDF use the same file. Third-party images name their owner in the caption and aren't covered by the content *licence*.

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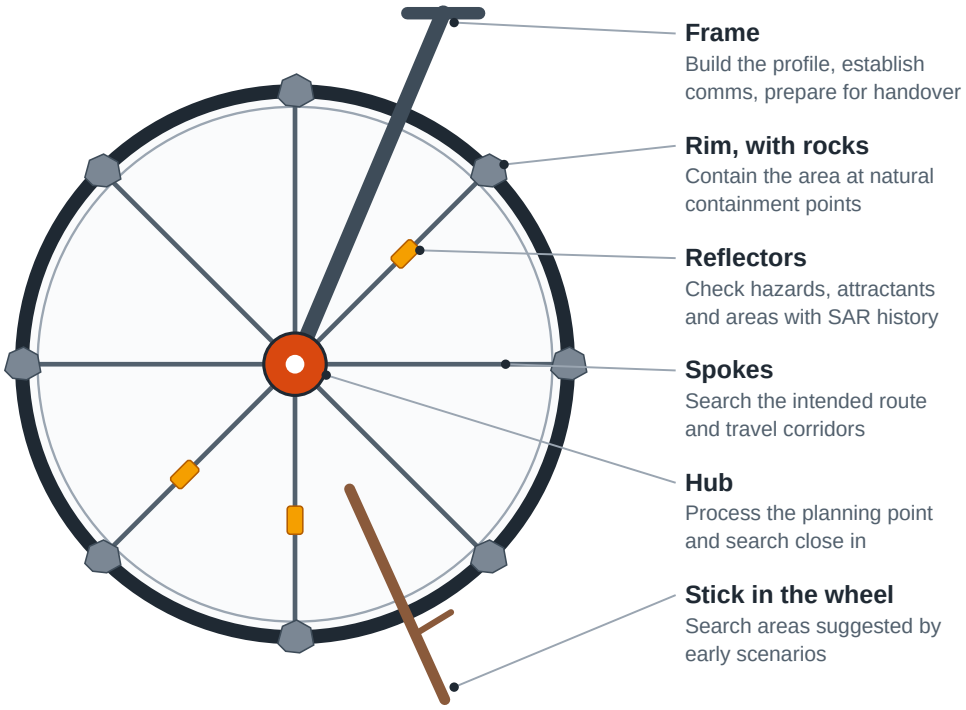


Figure 6.3: Each part of the wheel is an initial response objective (*CON-14*). A schematic only: no distances are shown, and attribution of the model is still being checked.

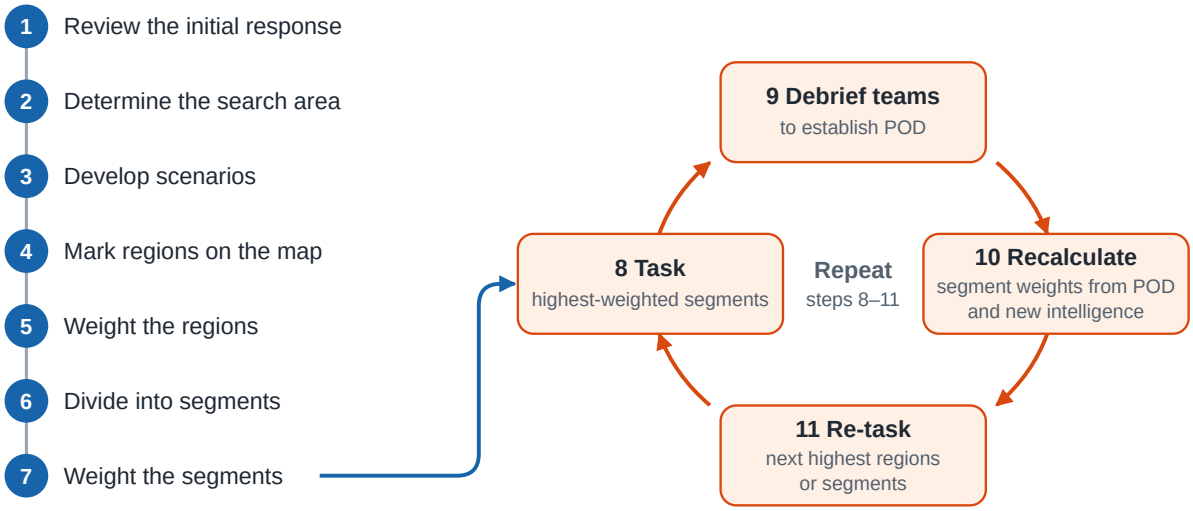


Figure 6.4: Steps 1–7 set up the search; steps 8–11 repeat (*CON-16*, *PRC-02*).



Figure 6.5: Logo of Land Search and Rescue New Zealand (PNG). Shown only as an example of a raster image; it doesn't imply endorsement.

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Figure 6.6: Logo of New Zealand Search and Rescue (PNG). Shown only as an example of a raster image; it doesn't imply endorsement.

CHAPTER

SEVEN

GLOSSARY

Each term is defined once here, and pages link to it with `{term}`POD``. Terms follow *CIMS* usage where CIMS defines them.

ACH
: Analysis of Competing Hypotheses. A structured method for testing scenarios against evidence, favouring the scenario least contradicted by it.

CIMS
: Coordinated Incident Management System. New Zealand's common framework for managing incidents across agencies.

CIS
: Close-In Search. A search of the immediate area around the initial planning point for the subject, clues and direction of travel.

COP
: Common Operating Picture. The single shared display of the incident: plans, maps, intelligence, resources and timelines.

DOT
: Direction of Travel.

ESP
: Extended Search Planning. The phase that follows the initial response when the subject has not been found.

FLO
: Family Liaison Officer.

IAP
: Incident Action Plan. The plan for one operational period: mission, objectives, strategies, tasks, factors and safety.

ICP
: Incident Control Point. The location from which the incident is controlled. "Information collection plan" is always written in full, never abbreviated.

IMT
: Incident Management Team.

IPP
: Initial Planning Point. The point distances are measured from, usually the LKP or PLS.

IR
: Initial Response. The first phase of a search, driven by pre-determined, statistically based actions.

LKP
: Last Known Point. The last location the subject is known to have been, from physical evidence.

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LPB
: Lost Person Behaviour. Statistical data on how people in different subject categories behave and where they are found.

MBO
: Management by Objectives.

MMISSI TD
: Mnemonic for the initial response process: missing person report, map, information collection plan, scenario, search urgency, incident action plan, task, debrief.

MOST
: Mission, Objectives, Strategies, Tasks. The structure of an IAP.

MPR
: Missing Person Report.

NZSAR
: New Zealand Search and Rescue. The NZSAR Council gives national strategic governance of the SAR system.

OSC
: On-Scene Coordinator.

PLS
: Point Last Seen. The last location where someone saw the subject.

POA
: Probability of Area. The likelihood that the subject is in a given region or segment.

POD
: Probability of Detection. The likelihood that a search of an area would have found the subject if they were there.

POS
: Probability of Success. POA × POD for a searched area.

RCCNZ
: Rescue Coordination Centre New Zealand. Coordinating Authority for Category II operations.

SAD
: Search Area Determination. Setting the boundary of the search area.

SAR
: Search and Rescue.

SAREX
: Search and rescue exercise.

SAROP
: Search and rescue operation.

SEE
: Search Effort Evaluation. The team debrief that establishes coverage and POD.

SET
: Speed, Efficiency, Thoroughness. Tells a team which matters most for its task.

SMART
: Specific, Measurable, Achievable, Relevant, Time-bound. Criteria for objectives.

SMS
: Safety Management System.

SOP
: Standard Operating Procedure.

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CHAPTER

EIGHT

TRACEABILITY STATUS

Generated 2026-10-18 from 44 items: 15 principles, 17 concepts, 6 guidelines, 3 processes, 3 forms.

8.1 Needs attention

Status	Item	Detail
warning	*PRN-15 Iwi and community partnerships*	no concept derives from this principle yet
stale	*CON-02 Risk management and safety*	written against PRN-13 v1; PRN-13 is now v2
overdue	*GDL-04 Rescue phase: known location*	last reviewed 2025-06-30

8.2 Derivation tree

- *PRN-01 Unified and consistent approach*, v1
  - *CON-01 Hierarchical framework of the doctrine*, v1
- *PRN-02 Adherence to a national management framework (CIMS)*, v1
  - *CON-04 Notification and awareness*, v1
    - *PRC-03 Notification and on-scene coordination*, v1
  - *CON-06 Incident management teams and CIMS*, v1
    - *GDL-01 Managing an IMT: key actions*, v1
- *PRN-03 Fostering partnerships and relationships*, v1
  - *CON-05 Leadership*, v1
  - *CON-06 Incident management teams and CIMS*, v1
    - *GDL-01 Managing an IMT: key actions*, v1
- *PRN-04 Importance of regional collaboration*, v1
  - *CON-01 Hierarchical framework of the doctrine*, v1
- *PRN-05 Proactive planning and readiness*, v1
  - *CON-03 Pre-operational readiness plans*, v1
  - *CON-13 Incident action plans*, v1
    - *GDL-03 Handovers, demobilisation and operational debriefs*, v1
      - *FRM-03 IMT summary checklist*, v1

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- *PRN-06 Effective and supportive leadership*, v1
  - *CON-05 Leadership*, v1
- *PRN-07 The critical role of communications*, v1
  - *CON-10 Communications*, v1
- *PRN-08 Evidence-based and intelligence-led operations*, v1
  - *CON-08 Maths in SAR*, v1
  - *CON-09 Lost and missing person behaviour*, v1
  - *CON-11 Evidence-based and intelligence-led operations*, v1
    - *GDL-01 Managing an IMT: key actions*, v1
  - *CON-17 Search theory: POA × POD = POS*, v1
    - *GDL-05 Extended search planning phase*, v1
      - *FRM-03 IMT summary checklist*, v1
      - *PRC-02 Extended search planning cycle*, v1
        - *FRM-01 Team debrief (search effort evaluation)*, v1
        - *FRM-02 Scenario weighting sheet*, v1
    - *GDL-06 Suspension*, v1
      - *FRM-03 IMT summary checklist*, v1
- *PRN-09 Search urgency*, v1
  - *CON-04 Notification and awareness*, v1
    - *PRC-03 Notification and on-scene coordination*, v1
  - *CON-15 The initial response process*, v1
    - *GDL-02 Initial response phase*, v1
      - *FRM-03 IMT summary checklist*, v1
      - *PRC-01 Initial response (MMISSI TD)*, v1
        - *FRM-01 Team debrief (search effort evaluation)*, v1
    - *PRC-01 Initial response (MMISSI TD)*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
- *PRN-10 Structured information management*, v1
  - *CON-11 Evidence-based and intelligence-led operations*, v1
    - *GDL-01 Managing an IMT: key actions*, v1
- *PRN-11 Systematic and methodical approach: POA × POD = POS*, v1
  - *CON-07 Phases of a land search and rescue operation*, v1
    - *GDL-04 Rescue phase: known location*, v1
    - *GDL-06 Suspension*, v1
      - *FRM-03 IMT summary checklist*, v1
  - *CON-08 Maths in SAR*, v1

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- *CON-12 Management by objectives*, v1
  - *GDL-01 Managing an IMT: key actions*, v1
  - *GDL-03 Handovers, demobilisation and operational debriefs*, v1
    - *FRM-03 IMT summary checklist*, v1
- *CON-13 Incident action plans*, v1
  - *GDL-03 Handovers, demobilisation and operational debriefs*, v1
    - *FRM-03 IMT summary checklist*, v1
- *CON-14 The Bike Wheel model*, v1
  - *GDL-02 Initial response phase*, v1
    - *FRM-03 IMT summary checklist*, v1
    - *PRC-01 Initial response (MMISSI TD)*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
- *CON-15 The initial response process*, v1
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      - *FRM-01 Team debrief (search effort evaluation)*, v1
  - *PRC-01 Initial response (MMISSI TD)*, v1
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  - *GDL-05 Extended search planning phase*, v1
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    - *PRC-02 Extended search planning cycle*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
      - *FRM-02 Scenario weighting sheet*, v1
  - *PRC-02 Extended search planning cycle*, v1
    - *FRM-01 Team debrief (search effort evaluation)*, v1
    - *FRM-02 Scenario weighting sheet*, v1
- *CON-17 Search theory: POA × POD = POS*, v1
  - *GDL-05 Extended search planning phase*, v1
    - *FRM-03 IMT summary checklist*, v1
    - *PRC-02 Extended search planning cycle*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
      - *FRM-02 Scenario weighting sheet*, v1
  - *GDL-06 Suspension*, v1
    - *FRM-03 IMT summary checklist*, v1

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- *PRN-12 The principle of simplicity*, v1
  - *CON-15 The initial response process*, v1
    - *GDL-02 Initial response phase*, v1
      - *FRM-03 IMT summary checklist*, v1
      - *PRC-01 Initial response (MMISSI TD)*, v1
        - *FRM-01 Team debrief (search effort evaluation)*, v1
    - *PRC-01 Initial response (MMISSI TD)*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
- *PRN-13 Prioritising safety*, v2
  - *CON-02 Risk management and safety*, v1 (stale)
- *PRN-14 Continuous improvement*, v1
  - *CON-01 Hierarchical framework of the doctrine*, v1
- *PRN-15 Iwi and community partnerships*, v1

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CHAPTER

NINE

HOW THIS DOCTRINE IS MAINTAINED

9.1 Two review tiers

Tier	Folders	To merge a change
Doctrine	Parts 1, 2, 3.1, 3.2, glossary, shared snippets	Approval from a doctrine owner (sector review group)
Guidance	3.3–3.6, forms	One approval from any reviewer

Doctrine changes slowly and needs agreement; guidance and forms change often and need light review. The split is set in `.github/CODEOWNERS`.

9.2 Items and derivation

Principles, concepts, guidelines, processes and forms are **items**. Each item page starts with a short block of metadata:

```yaml
---
id: GDL-05
type: guideline
title: Extended search planning phase
version: 1
derives_from:      # parent ID: parent version this was written against
  CON-16: 1
  CON-17: 1
owner: doctrine-group
last_reviewed: 2026-10-01
---
```

Allowed links run down the hierarchy: principle → concept → guideline → process → form. A guideline can't derive straight from a principle.

When an item's meaning changes, its author bumps `version`. Every child written against the old version is then listed as **stale** on the *Traceability status* page until someone checks it and updates its `derives_from` number. The tooling confirms that someone *looked*; it can't judge whether the child is still right.

9.3 Adding a page

Each type of item has its own folder, ID prefix and allowed parents:

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Type	Folder	ID	Derives from
Principle	`docs/part-2/`	`PRN-`	nothing
Concept	`docs/part-3/3-1-concepts/`	`CON-`	principles
Guideline	`docs/part-3/3-3-guidelines/`	`GDL-`	concepts
Process	`docs/part-3/3-4-processes/`	`PRC-`	concepts or guidelines
Form	`docs/forms/`	`FRM-`	guidelines or processes

1. On GitHub, open the folder and choose **Add file → Create new file**.
2. Name the file after the next unused ID in lower case, then a short name: `gdl-07-night-operations.md`. Pages are listed in filename order, so always use two digits.
3. Start the file with its metadata and a heading. Under `derives_from`, give each parent's ID and its current version, shown in the box at the top of its page.

```
---
id: GDL-07
type: guideline
title: Night operations
version: 1
derives_from:
  CON-16: 1
owner: doctrine-group
last_reviewed: 2026-10-17
---

# GDL-07 Night operations

The text of the guideline.
```

4. Choose **Commit changes** and propose the change. GitHub opens a pull request, the checks run and a preview of the site is built.

The new page appears in its folder's contents and the index without further edits. Three optional fields: `requires_children: [form]` flags the item until a form derives from it, `review_days` changes the 365-day review interval, and `index: [incident controller]` adds index entries for topics the title doesn't name.

9.4 Checks on every change

Check	Result
ID already used, or its prefix doesn't match the type	Fails
Parent ID doesn't exist, or wrong parent type	Fails
Child claims a newer parent version than exists	Fails
Item needs children and has none	Warning
Child written against an older parent version	Listed as stale
Parent text changed but version not bumped	Warning
`last_reviewed` older than the review interval	Listed as overdue

Each pull request also gets a **Traceability** comment listing the findings it adds or resolves, and each release's changelog records the totals at release.

9.5 Say it once

- Repeated text, such as the *POD* scale, lives in one snippet file and is included on every page that needs it (four pages here).
- Acronyms are defined once in the *Glossary*. Pages link to them with `{term}`POD``, so the glossary and the genindex list every page that uses each one.
- The organisation name comes from one setting, so it reads “LandSAR” everywhere.

9.6 What the demo content shows

Text is condensed from draft v0.1, numbered as in the draft. Author notes, names and statistics still being verified are left out. Three states are deliberate, so the *Traceability status* page has something to show:

State	Item	Why
Stale	*CON-02 Risk management and safety*	Written against PRN-13 v1; PRN-13 is now v2
Coverage gap	*PRN-15 Iwi and community partnerships*	No concept derives from it yet
Overdue review	*GDL-04 Rescue phase: known location*	Last reviewed more than a year ago

9.7 Releases

Each release has a version number like `v1.2.0`: edition, amendment and correction. A new edition means doctrine in Parts 1, 2 or 3.1–3.2 changed meaning; an amendment means guidance, processes or forms changed; a correction changes wording only. Release candidates such as `v2.0.0-rc.1` go out for consultation before the final version. Each release stays online at its own address, next to the latest release and the working draft, with a PDF, and the *Changelog* lists what changed and who contributed. The version menu at the top of the sidebar switches between them. Operational reviews cite the version number shown at the top of every page. Between releases, the working draft shows the last release and the number of changes since, such as *v0.2.2-demo + 3 changes*, and its front page gives the commit to cite.

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CHAPTER

TEN

CHANGELOG

10.1 v0.2.3-demo — 2026-10-18

10.1.1 Site and tooling

- Older versions on the site now show the version menu and a banner pointing to the latest release. (#24, Tobias Schmidt)
- The working draft now shows its version as the last release plus the number of changes since, such as “v0.2.2-demo + 3 changes”. (#26, Tobias Schmidt)

10.1.2 Needs review

- **CON-02** written against PRN-13 v1; PRN-13 is now v2

Traceability at release: 1 warning, 1 stale, 1 overdue.

Contributors: Tobias Schmidt

10.2 v0.2.2-demo — 2026-10-18

10.2.1 Doctrine

- New pages appear in their part’s contents automatically, and “How this doctrine is maintained” explains how to add one. (#18, Tobias Schmidt)

10.2.2 Site and tooling

- The site now keeps every release at its own address, with a version menu, next to the latest release and the working draft. (#20, Tobias Schmidt)

10.2.3 Needs review

- **CON-02** written against PRN-13 v1; PRN-13 is now v2

Traceability at release: 1 warning, 1 stale, 1 overdue.

Contributors: Tobias Schmidt

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10.3 v0.2.1-demo — 2026-10-17

10.3.1 Guidance and forms

- Added a Figures page with diagrams of the doctrine levels, the phases of an operation, the Bike Wheel model and the extended search planning cycle, plus two logos as examples of raster images. (#13, Tobias Schmidt)

10.3.2 Site and tooling

- Each pull request now gets a comment showing how it changes traceability, and each release records its traceability totals. (#14, Tobias Schmidt)
- The release pull request now updates itself after every merge, so it always shows what the next release will contain. (#17, Tobias Schmidt)

10.3.3 Needs review

- **CON-02** written against PRN-13 v1; PRN-13 is now v2

Traceability at release: 1 warning, 1 stale, 1 overdue.

Contributors: Tobias Schmidt

10.4 v0.2.0-demo

- Condensed content from draft v0.1 for Parts 1, 2, 3.1 and 3.3–3.6: 15 principles, 17 concepts, 6 guidelines, 3 processes and 3 forms, numbered as in the draft.
- Expanded glossary; POD scale shown as drafted, in one shared snippet.
- Demonstration states: stale CON-02, coverage gap PRN-15, overdue GDL-04.

10.5 v0.1.0-demo

- Repository structure for Parts 1–3 and forms.
- One traced example thread: evidence-based decisions → POA/POD/POS → recording POD → search period cycle → team debrief form.
- Deliberate demonstration states: one stale item (CON-03), one coverage gap (PRN-04), one overdue review (GDL-01).

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CHAPTER

ELEVEN

CONTRIBUTORS

Everyone who has contributed to this doctrine. People are added when a release includes a pull request they wrote or co-wrote; add anyone who contributed another way, such as through workshops or review comments, by hand.

- Tobias Schmidt

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CHAPTER

TWELVE

LICENCE AND AUTHORITY

Only doctrine published by Land Search and Rescue New Zealand (LandSAR) is authoritative. Check the version on the front page against the latest release before relying on a copy.

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